

Module 5 Activity 4: Working with Zonal Functions

Purpose:

Incorporate the **arcpy.sa** module to work with neighborhood functions to generate new tables and analyze the results.

Instructions:

Any Python environment will work for this exercise.

Download the data **Module 5 Activity 4** and save the data to your working directory.

Part 1 – Zonal Statistics as Table

1. Import the **arcpy** module
2. Import the **arcpy.sa** module
3. Check out the spatial analyst extension so it can be used
4. Add a **print** statement to the **CheckOutExtension** function to notify the user that it is ready for use
5. Set the workspace and overwrite outputs

There will be additional blank **print** statements used throughout the script. This is to make the results easy to read. They won't be part of the process steps outlined in this activity and are completely optional.

```
Python
#import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\GEOS456\ZonalStats.gdb"
```

6. Create a list of raster datasets present in the geodatabase
7. Describe the raster properties of each raster
 - Format
 - Cell size
 - Spatial reference
8. Create a list of feature classes present in the geodatabase

```
Python
#import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = "C:\GEOS456\ZonalStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#print statement to show what rasters are inside the geodatabase
print ("The following rasters are present in the gdb:")
#start a for loop to iterate through the list
for rasters in rasterlist:
    desc = arcpy.Describe(rasters)
    print (rasters)
    print ("The format is:", desc.format)
    print ("The cell size is:", desc.meanCellWidth)
    print ("The coordinate system is:", desc.spatialReference.name)
    print ("")
#get a list of feature classes in the geodatabase
featurelist = arcpy.ListFeatureClasses()
for features in featurelist:
    print (features)
    print ("")
```

The next steps will use the **Zonal** toolset to generate a new raster with the **Zonal Statistics as Table** tool to generate a table showing the mean elevation for each land cover species

9. Define a variable for each required parameter of the **Zonal Statistics as Table** tool
10. Apply the **Zonal Statistics as Table** tool
11. Use a **print** statement to tell the user that the table has been created

```
Python
#import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = r"C:\GEOS456\ZonalStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#print statement to show what rasters are inside the geodatabase
print ("The following rasters are present in the gdb:")
#start a for loop to iterate through the list
for rasters in rasterlist:
    desc = arcpy.Describe(rasters)
    print (rasters)
    print ("The format is:", desc.format)
    print ("The cell size is:", desc.meanCellWidth)
    print ("The coordinate system is:", desc.spatialReference.name)
    print ("")
#get a list of feature classes in the geodatabase
featurelist = arcpy.ListFeatureClasses()
for features in featurelist:
    print (features)
    print ("")
#define the input parameters of the zonal stats as table tool
InZoneData = "Landcover"
ZoneField = "Species_gr"
InValueRaster = "DEM"
OutTable = "Landcvt_Mean_elev"
#create a table using Zonal Statistics as Table from the zonal toolset
#set the stats type to MEAN
OutZoneTable = ZonalStatisticsAsTable(InZoneData,ZoneField,InValueRaster,OutTable,"", "MEAN")
#print a statement that informs the user that the table has been created
print ("Zonal Stats table has been created with MEAN stats type.")
print ("")
```

The table has been generated. Next, we will create a list of the fields and access the values from the list.

12. Use a **DATA ACCESS** search cursor to retrieve information from the **Species_gr** and the **MEAN** fields in the table
13. Use a **for** loop to iterate through the fields
14. Use a **print** statement to show the results
15. Check the spatial analyst extension back in to the license manager

```
Python
# import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
# check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") # blank print for ease of reading
# set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = r"C:\GEOS456\ZonalStats.gdb"
# get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
# print statement to show what rasters are inside the geodatabase
print ("The following rasters are present in the gdb:")
# start a for loop to iterate through the list
for rasters in rasterlist:
    desc = arcpy.Describe(rasters)
    print (rasters)
    print ("The format is:", desc.format)
    print ("The cell size is:", desc.meanCellWidth)
    print ("The coordinate system is:", desc.spatialReference.name)
    print ("")
# get a list of feature classes in the geodatabase
featurelist = arcpy.ListFeatureClasses()
for features in featurelist:
    print (features)
    print ("")
# define the input parameters of the zonal stats as table tool
InZoneData = "Landcover"
ZoneField = "Species_gr"
InValueRaster = "DEM"
OutTable = "Landcvr_Mean_elev"
# create a table using Zonal Statistics as Table from the zonal toolset
# set the stats type to MEAN
OutZoneTable = ZonalStatisticsAsTable(InZoneData, ZoneField, InValueRaster, OutTable, "", "MEAN")
# print a statement that informs the user that the table has been created
print ("Zonal Stats table has been created with MEAN stats type.")
print ("")
# use a DATA ACCESS search cursor to retrieve values from the attribute fields
SCursor = arcpy.da.SearchCursor(OutTable, ["Species_gr", "MEAN"])
# use a for loop to iterate through the fields
for row in SCursor:
    # print the results for easy reading
    print ("Species: ", row[0], "|", "MEAN Elevation: ", row[1])
# check out the spatial analyst extension
arcpy.CheckInExtension("Spatial")
print ("")
print ("Spatial Analyst returned to license manager.")
```

16. Run the script and examine you results

```
Python
Spatial Analyst extension checked out and ready to use.

The following rasters are present in the gdb:
DEM
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Hillshade
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Landcover
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Study_Area

Streams

Fire_Poly

Zonal Stats table has been created with MEAN stats type.

Species: MIXED CONIFER - FIR | MEAN Elevation: 1968.3276439387182
Species: MONTANE CHAPARRAL (Mid elev.) | MEAN Elevation: 1874.9839142091166
Species: MIXED CONIFER - PINE | MEAN Elevation: 1963.3350847457643
Species: CANYON LIVE OAK | MEAN Elevation: 1726.9208907119855
Species: RED FIR | MEAN Elevation: 2314.281600588525
Species: WATER | MEAN Elevation: 1481.791304347826
Species: HUCKLEBERRY OAK (High elev.) | MEAN Elevation: 2300.2179487179496
Species: INTERIOR LIVE OAK | MEAN Elevation: 1718.6377551020407
Species: PONDEROSA PINE | MEAN Elevation: 1728.5386100386106
Species: BARREN | MEAN Elevation: 1875.7882352941167
Species: WET MEADOWS GRASS/SEDGE/RUSH | MEAN Elevation: 2265.281904761905
Species: MIXED CHAPARRAL (Low elev.) | MEAN Elevation: 1501.2304403591284
Species: MP W/ GIANT SEQUOIA | MEAN Elevation: 1880.1425470332852
Species: BIGTREE (giant sequoia) | MEAN Elevation: 1893.138297872341
Species: JEFFREY PINE | MEAN Elevation: 1733.3802497162305
Species: LODGEPOLE PINE | MEAN Elevation: 1967.9688359303373
Species: ANNUAL GRASS/FORBS | MEAN Elevation: 1845.47266881029

Spatial Analyst returned to license manager.
```

Next Steps:

Run the same process as above using different statistics types. Compare your results. The **Input raster or feature zone data** in the tool can be either raster or vector. Repeat the steps above using the **Streams** feature class to determine the mean elevation of each river class.

Part 2 – Tabulate Area

The tabulate area tool returns tabular information about the area of a raster dataset. It is the best method to generate areas of a raster.

1. Repeat **Steps 1-8** from **Part 1**

2. Specify a variable for each required parameter of the **Tabulate Area** tool
3. Apply the **Tabulate Area** tool
4. Use a **print** statement to tell the user that the table has been created
5. Use a **DATA ACCESS** search cursor to retrieve information from the following land cover types:
 - **PONDEROSA_PINE**
 - **RED_FIR**
6. Use a **for** loop to iterate through the fields
7. Use a **print** statement to show the results
8. Check the spatial analyst extension back in to the license manager

```

Python
#import arcpy and spatial analyst modules
import arcpy
from arcpy import env
from arcpy.sa import *
#check out the spatial analyst extension
arcpy.CheckOutExtension("Spatial")
print ("Spatial Analyst extension checked out and ready to use.")
print ("") #blank print for ease of reading
#set the overwrite outputs and workspace
env.overwriteOutput = True
env.workspace = r"C:\GEOS456\ZonalStats.gdb"
#get a list of the rasters in the geodatabase
rasterlist = arcpy.ListRasters()
#print statement to show what rasters are inside the geodatabase
print ("The following rasters are present in the gdb:")
#start a for loop to iterate through the list
for rasters in rasterlist:
    desc = arcpy.Describe(rasters)
    print (rasters)
    print ("The format is:", desc.format)
    print ("The cell size is:", desc.meanCellWidth)
    print ("The coordinate system is:", desc.spatialReference.name)
    print ("")
#get a list of feature classes in the geodatabase
featurelist = arcpy.ListFeatureClasses()
for features in featurelist:
    print (features)
    print ("")
#define the input parameters of the tabulate area tool
InZoneData = "Fire_Poly"
ZoneField = "FireNum"
InValueRaster = "Landcover"
ClassField = "Species_gr"
OutTable = "Fire_Area"
#create a table using the tabulate area tool
OutAreaTable = TabulateArea(InZoneData,ZoneField,InValueRaster,ClassField,OutTable)
#print a statement that informs the user that the table has been created
print ("Tabulate Area table has been created.")
print ("")
#use a DATA ACCESS search cursor to retrieve values from the attribute fields
SCursor = arcpy.da.SearchCursor(OutTable,["FIRENUM","PONDEROSA_PINE","RED_FIR"])
#use a for loop to iterate through the fields
for row in SCursor:
    #print the results for easy reading
    print ("Fire Number: ", row[0], "|", "Ponderosa Pine Area (sq/m): ", row[1], "|", "Mixed Conifer Area (sq/m): ", row[2])
#check out the spatial analyst extension
arcpy.CheckInExtension("Spatial")
print ("")
print ("Spatial Analyst returned to license manager.")

```

9. Run the script and examine you results

```
Python
Spatial Analyst extension checked out and ready to use.

The following rasters are present in the gdb:
DEM
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Hillshade
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Landcover
The format is: FGDBR
The cell size is: 30.0
The coordinate system is: UTM

Study_Area

Streams

Fire_Poly

Tabulate Area table has been created.

Fire Number: 1 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 0.0
Fire Number: 2 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 2997000.0
Fire Number: 3 | Ponderosa Pine Area (sq/m): 274500.0 | Mixed Conifer Area (sq/m): 0.0
Fire Number: 4 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 0.0
Fire Number: 5 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 0.0
Fire Number: 6 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 82800.0
Fire Number: 7 | Ponderosa Pine Area (sq/m): 900.0 | Mixed Conifer Area (sq/m): 3511800.0
Fire Number: 8 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 900.0
Fire Number: 9 | Ponderosa Pine Area (sq/m): 0.0 | Mixed Conifer Area (sq/m): 0.0

Spatial Analyst returned to license manager.
```

Next Steps:

Use the **Landcover** raster as both the **InZoneData** and the **InValueRaster** to generate the area of each of the species.